

Visa Acceptance Solutions for Salesforce B2C Commerce REST v2.0.0



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Salesforce B2C Commerce

The Visa Acceptance Solutions cartridge for Salesforce B2C Commerce enables merchants to connect their Salesforce B2C Commerce store to the Visa Acceptance Platform for payment processing.

Supported Features

The Visa Acceptance Solutions cartridge for Salesforce B2C Commerce supports multiple payment methods and features.

Payment Methods

Through Unified Checkout, a single integration supports the following payment methods

- Credit/debit cards
- Apple Pay
- Google Pay
- Paze
- PayPal
- Venmo
- Click to Pay
- eCheck

Through the default Salesforce payment form calling Visa Acceptance Solutions API's directly, the following payment methods are supported:

- Credit/debit cards
- Apple Pay

Security and Fraud Management

- Payer Authentication/3-D Secure
- Token Management Service
- Decision Manager and Fraud Management Essentials

Additional Services

- Delivery Address Verification
- Tax Calculation
- Meta-key

Methods are also exposed to process authorization reversal, capture, and refund.

Supported Versions

The Visa Acceptance Solutions cartridge is compatible with specific versions of Salesforce Storefront Reference Architecture (SFRA).

Compatibility

The Visa Acceptance Solutions cartridge is compatible with Salesforce SFRA version 7.0 and earlier.

Visa Acceptance Solutions Prerequisites

Before implementing the Visa Acceptance Solutions cartridge, ensure that you have the required and optional Visa Acceptance Solutions products configured.

Mandatory

You must have a [REST Shared Secret Key Pair](#).

Optional

These products are optional. If you choose to use any of these products, they must be enabled and configured for your Merchant ID:

- Decision Manager
- Fraud Management Essentials
- Network Tokens

As of version 2.0.0 for Unified Checkout, accepted card brands, payment methods, Payer Authentication for 3-D Secure, fraud screening, and payment action is configured and enabled in the Business Center. For more information, see the [Unified Checkout Configuration Guide](#).

If you are going to use Unified Checkout, you must have a [Response MLE Certificate](#)

Install Visa Acceptance Solutions Salesforce B2C Commerce

Download and install the Visa Acceptance Solutions cartridge for Salesforce B2C Commerce.

Before beginning installation, ensure that you have:

- Access to your Salesforce B2C Commerce instance.
 - Node.js installed on your development machine.
 - Appropriate IDE, such as VSCode with the Prophet Debugger extension.
1. Download the Visa Acceptance Solutions cartridge for Salesforce B2C Commerce from the ISV Integration Toolkits section on [GitHub](#).
 2. Set up your workspace by creating a folder named `Visa Acceptance Solutions` in your Salesforce workspace and copy the downloaded cartridges, `int_cybs_sfra`, `int_cybs_sfra_base`, and `bm_cybs_sfra` to the workspace.
If the project's base path is different from the one available in `package.json`, open the file `/package.json` and modify the `paths.base` value to point to your `app_storefront_base` cartridge. This path is used by the JS and SCSS build scripts.
 3. Configure the IDE.
If you use VSCode, install the Prophet Debugger extension and include these lines in `dw.json`:

```
{
  "hostname": "your-sandbox-hostname.demandware.net",
  "username": "yourlogin",
  "password": "yourpwd",
  "version": "version_to_upload_to",
  "cartridge": [
    "int_cybs_sfra",
    "int_cybs_sfra_base",
    "bm_cybs_sfra",
    "app_storefront_base",
    "modules"
  ]
}
```

If you are using a different IDE, refer to the respective guide to set up your workspace.

Build and Upload Code

1. Install the Node.js dependencies in the Visa Acceptance Solutions folder.
2. Install *sgmf-scripts* and *copy-webpack-plugin* with this command:

```
npm install sgmf-scripts && npm install copy-webpack-plugin
```

3. Compile JS and SCSS with this command:

```
npm run compile:js && npm run compile:scss
```

4. Upload the code to the Salesforce Commerce Cloud instance:

```
npm run uploadCartridge
```

The Visa Acceptance Solutions cartridge is now installed and ready for configuration in your Salesforce B2C Commerce environment.

Configure Visa Acceptance Solutions for Salesforce B2C Commerce

Configure the Visa Acceptance Solutions cartridge in the Salesforce B2C Commerce Business Manager.

After installation, configure the cartridge through the Salesforce Business Manager to enable payment processing capabilities.

Base Configuration

1. Set up the cartridge path in Salesforce Business Manager by going to **Administration > Sites > Manage Sites > [yourSite] > Settings**.
For Cartridges, enter *bm_cybs_sfra:int_cybs_sfra:int_cybs_sfra_base:app_storefront_base* and click **Apply**.

Upload Metadata

2. Go to the folder *Visa Acceptance Solutions/metadata/payments_metadata/sites/*.
3. Rename the folder *yourSiteID* with your site ID from Salesforce Business Manager.
This can be found by going to **Administration > Sites > Manage Sites**.

4. Zip the *payments_metadata* folder.
5. Go to **Administration > Site Development > Site Import & Export** and upload the *payments_metadata.zip* file.
6. Import the uploaded zip file.

Upon successful import, this metadata is created:

- Site Preferences: VisaAcceptance_Core, VisaAcceptance_SecureIntegrationConfiguration, VisaAcceptance_SalesforceDefaultAcceptance_Configuration, VisaAcceptance_ApplePay, VisaAcceptance_Tokenization, VisaAcceptance_DeviceFingerprint, VisaAcceptance_DeliveryAddressVerification, VisaAcceptance_TaxConfiguration, VisaAcceptance_MLE
- Service: PaymentHttpService
- Payment Processor
- Payment Method
- Jobs: Payment : Decision Manager Order Update, Payment: Refresh Payment Status



Important: As of version 2.0.0, Payer Authentication (including SCA), Decision Manager, and Transaction Type are now grouped under Salesforce Default Acceptance Configuration, and apply to the Salesforce default card form (Direct API) only. Standalone Google Pay preferences have been removed.

Minimum Configuration

7. Configure the Visa Acceptance Core preferences by going to **Merchant Tools > Site Preferences > Custom Preferences > Visa Acceptance Cartridge Configuration** and setting these configuration parameters:
 - **Enable Visa Acceptance Cartridge:** Enable the cartridge.
 - **Visa Acceptance Solutions Merchant ID:** The transacting Merchant ID assigned to you.
 - **Visa Acceptance Solutions REST Key Id:** The key from your REST API shared secret key.
 - **Visa Acceptance Solutions REST Secret Key:** The shared secret from your REST API shared secret key.
 - Commerce Indicator:
 - For eCommerce transactions, use [internet](#).
 - When you are using the store for call center transactions only, use [MOTO](#).

- **Enable Visa Acceptance Test Endpoints:** Disabled by default. This enables the capture and authorization reversal endpoints ([ServiceFrameworkTest](#) controller). Use only in a sandbox environment.
- **Enable Meta Key:** Enable Meta Key for portfolio/account-level key management (*optional*).
- **Meta Key Portfolio Merchant ID:** The Visa Acceptance Solutions portfolio/account Merchant ID that owns the Meta Key (*optional*).

Services

The target endpoints need to be set in your Salesforce B2C Commerce environment.

8. Configure the services by going to **Merchant Tools > Operations > Services > Payment Credentials** and enter the following URLs:

- **Environment:** Test **URL:** <https://apitest.visaacceptance.com>
- **Environment:** Production **URL:** <https://api.visaacceptance.com>

Payment Capture Method

9. Go to **Merchant Tools > Ordering > Payment Methods**, select [CREDIT_CARD](#), and verify that the payment processor is [PAYMENTS_CREDIT](#).

10. Go to **Merchant Tools > Site Preferences > Custom Preferences > Secure Integration Configuration**.

Choose from

- Unified Checkout
- Direct API (default Salesforce payment form)

Unified Checkout might qualify you for PCI-DSS SAQ:A because the card number is collected in secure fields and never resides on your server.

If you need access to the card number, select [Direct API](#). This option increases your PCI burden.

If you select Unified Checkout, please see [Unified Checkout Configuration \(on page 10\)](#)

The Visa Acceptance Solutions cartridge is now configured with basic settings.

Optional Configuration

Additional optional configurations can be applied based on your specific requirements.

Unified Checkout Configuration

Configuration requirements and options for Unified Checkout.

There are required and optional configuration options for Unified Checkout.

Unified Checkout Digital/Alternative Payment Methods Payment Processor Mapping

If you want to offer Digital or Alternative Payment Methods via Unified Checkout, ensure each payment method has the correct Payment Processor mapping.

Go To **Merchant Tools > Ordering > Payment Methods** and verify the following mappings:

- DW_APPLE_PAY - Payment Processor `PAYMENTS_APPLEPAY`.
- DW_GOOGLE_PAY - Payment Processor `PAYMENTS_GOOGLEPAY`.
- DW_PAZE - Payment Processor `PAYMENTS_PAZE`.
- PAYPAL - Payment Processor `PAYMENTS_PAYPAL`.
- VENMO - Payment Processor `PAYMENTS_VENMO`.
- CLICK_TO_PAY - Payment Processor `PAYMENTS_CLICK_TO_PAY`.
- BANK_TRANSFER - Payment Processor `BANK_TRANSFER` (for eCheck)

Additional transaction responses from Unified Checkout are delivered by webhook notifications and require a Message Level Encryption Response Certificate. Please see [Webhooks \(on page 13\)](#) and [Response Message Level Encryption \(on page 12\)](#) for setup steps.

Unified Checkout Optional Configuration

To set the following options, go to **Merchant Tools > Site Preferences > Custom Preferences > Secure Integration Configuration**.

1. Set the **Checkout Label** for Unified Checkout.

Choose your own checkout page label. You can choose up to 60 characters, and the default value is "Secure Payments powered by Visa Acceptance Solutions"

2. Set the **Enable Express Pay** option.

Separates digital wallet payment methods from the main card capture widget.

3. Set the Unified Checkout **Version** option.

Pin a specific Unified Checkout version. Available for version 1.0 onwards. Leave this field blank for the latest version (recommended).

4. Set the **Card Prefix (BIN)** in Unified Checkout response option.

- a. **None:** the Card BIN is not returned
- b. **Six-digit:** the 6 digit BIN is returned
- c. **Eight-digit:** the 8 digit BIN is returned

5. Set the **Non-Wallet** Unified Checkout **Display Mode** option.

If Express Pay has been enabled, this applies to the non-express pay methods only.

- a. **Embedded:** the payment widget is displayed within the checkout page
- b. **Sidebar:** the payment widget appears on the side of the customer's browser.

Message-Level Encryption

Message Level Encryption (MLE) uses certificates to ensure that each message is securely encrypted and tied to the sender's verified identity, without needing to share secret keys in advance.



Important: Message-Level Encryption requires a Response MLE certificate to be created. The same certificate is used for **Request** and **Response** messages but need to be enabled separately. A REST Shared Secret is still required. See [Create a REST—API Response MLE Key](#) for steps on how to create a Response MLE certificate.

To enable Message Level Encryption for transaction processing, configure [Request Message Level Encryption \(on page 11\)](#) and [Response Message Level Encryption \(on page 12\)](#) first then go to **Merchant Tools > Site Preferences > Custom Preferences > Message-Level Encryption Configuration**. Set the field **Enable MLE** to **On**.

Request Message Level Encryption

Request Message Level Encryption can be setup using a Certificate Alias or through a single file using Salesforce IMPEX (import/export).



Important: IMPEX is WebDAV-accessible (authenticated, but read access is granted per directory and org-wide) and the .p12 contains your private key. The Certificate Alias setup imports the private key into Business Manager's protected store, where it cannot be read back or exported. Please consider this when deciding which method to use.

Follow these steps to setup Request Message Level Encryption.

1. Import Certificate with **Certificate Alias** method

a. Extract Certificate

- i. Extract certificate: convert the Response MLE p12 certificate to .crt format using this command:

```
openssl pkcs12 -in <key filename>.p12 -cacerts -nokeys -out <key filename>.crt
```

- ii. Open the converted file in a text editor and make sure *CyberSource_SJC_US* is the first certificate.

- b. Import certificate: Go to **Administration > Operations > Private Keys and Certificates** and import the extracted .crt. Make a note of the alias provided after the import is completed.

c. Go to **Merchant Tools > Site Preferences > Custom Preferences > Message-Level Encryption Configuration**:

- i. For the field **Request MLE Certificate Alias**, enter the alias used when importing the CyberSource_SJC_US certificate
- ii. Leave the field **Request MLE - P12 file path under IMPEX** blank.

2. Import Certificate with **Single File IMPEX** method

- a. Go to **Administration > Operations > Site Development > Import & Export** and upload the Response MLE p12 certificate

b. Go to **Merchant Tools > Site Preferences > Custom Preferences > Message-Level Encryption Configuration**:

- i. For the field **Request MLE - P12 file path under IMPEX**, enter the path under IMPEX where the certificate was uploaded - *IMPEX/src/* or */on/demandware.servlet/webdav/Sites/Impex/src/mle/*
- ii. Leave the field **Request MLE Certificate Alias** blank.

Response Message Level Encryption

Response Message Level Encryption is optional however, if you are using Unified Checkout, the Message Level Encryption certificate is required for the subscription to the Unified Checkout webhook response notification.

Follow these steps to setup Response Message Level Encryption.

1. Import certificate: Go to **Administration > Operations > Private Keys and Certificates** and import the p12 certificate. You will be asked to enter an **Alias** for the certificate and enter the **password** you set when creating the p12 certificate. Make a note of the alias provided after the import is completed.
2. Go to **Merchant Tools > Site Preferences > Custom Preferences > Message-Level Encryption Configuration**:
For the field **Private Key Alias for Response MLE and Webhooks**, enter the alias you set when importing the p12 certificate

Webhooks

The Visa Acceptance Platform sends asynchronous webhook notifications to your Salesforce B2C Commerce store for fraud decisions, and Unified Checkout event updates. Webhooks are the primary mechanism for keeping orders up to date. The Salesforce jobs remain available as a fallback

Webhook subscriptions are managed from a Business Manager page provided by the `bm_cybs_sfra` cartridge.

Prerequisite

1. Ensure `bm_cybs_sfra` cartridge is on the Business Manager cartridge path (see [Configure Visa Acceptance Solutions for Salesforce B2C Commerce \(on page 7\)](#))
2. Go to **Administration > Organization > Roles & Permissions** select the role, then go to **Business Manager Modules**, select the required context then enable **Webhook Manager** under **Visa Acceptance**.
3. For the Unified Checkout webhook, the Response Message Level Encryption key must be loaded and configured before registering the webhook. See [Response Message Level Encryption \(on page 12\)](#)

Webhook Groups and Events

The cartridge manages two webhook groups. Each notification is delivered to a controller route in the storefront cartridge.

Group	Product	Event Types	Notification Route
Fraud Management	Decision Manager	<code>risk.casemanagement.decision.accept</code> , <code>risk.casemanagement.decision.reject</code>	<code>WebhookNotification-dmNotification</code>
Fraud Management	Fraud Management Essentials	<code>risk.profile.decision.review</code> , <code>risk.casemanagement.d</code>	<code>WebhookNotification-dmNotification</code>

(continued)

Group	Product	Event Types	Notification Route
		<code>ecision.accept,</code> <code>risk.casemanagement.d</code> <code>ecision.reject</code>	
UC Webhook Events	Unified Checkout	<code>uc.orders.transaction</code> <code>results</code>	<code>WebhookNotification-p</code> <code>aymentNotification</code>

Subscribe to Webhooks

4. Enable the site preferences for the required webhooks, Decision Manager within **Salesforce Default Acceptance Configuration** and Unified Checkout within **Secure Integration Configuration**.
5. Click **Sync**. The cartridge will subscribe (or update) each enabled webhook based on the set Custom Site Preferences, and stores the subscription state.
6. Confirm each subscription has an **ACTIVE** status. If the status is **PENDING_REVIEW**, **INACTIVE**, or **SUSPENDED**, the subscription is not yet live. Wait up to 15 minutes and **Sync** again. Please contact Support if the status does not change to **ACTIVE**.

Subscription Storage

The webhook subscription state is stored in the organization-scoped custom object `VisaAcceptanceWebhookSubscription` (keyed by `ConfigId`), which records the `WebhookId`, `WebhookUrl`, `SecurityKey`, `Status`, `BaseUrl`, and `EgressPublicKey`. The object is managed by the Webhook Manager. You do not need to edit it manually.

Fallback: Salesforce Jobs

If webhook delivery is interrupted, these jobs can be used to reconcile the order state:

- **Payment: Decision Manager Order Update:** polls for Accept/Reject decisions on unconfirmed orders (fallback for Fraud Management webhooks).
- **Payment: Refresh Payment Status** (on-demand job): enter a comma-separated list of order numbers in the OrderNumbers parameter and use **Run Now** to refresh the Visa Acceptance payment status for those orders.

Configure Apple Pay Standalone

To offer Apple Pay outside of Unified Checkout, enable Apple Pay in your Salesforce B2C Commerce store.

Before you begin, complete [Integrating Apple Pay into Your System](#).

1. Configure Salesforce Business Manager.

a. From the menu, choose **Merchant Tools > Site Preferences > Apple Pay**.

b. Select the **Apple Pay Enabled?** check box.

c. Complete the Onboarding form:

- Ensure that the Apple Merchant ID and the Apple Merchant Name values you enter match the settings in your Apple account.
- Ensure that all other fields match your supported Visa Acceptance Solutions settings.
 - **Country Code:** Enter the country code for the location of your site. The country code is a two letter ISO 3166 country code (for example, [US](#)).
 - **Merchant Capabilities:** Check the box for 3-D Secure; leave the other fields unchecked.
 - **Supported Networks:** Select the types of payment you support; [Amex](#), [Mastercard](#), and [Visa](#) are supported by Visa Acceptance Solutions.
 - **Required Shipping Address Fields:** Select the fields that are required on the shipping form. Visa Acceptance Solutions recommends **Email, Name, Phone**, and **Postal Address**.
 - **Required Billing Address Fields:** Select **Name** and **Postal Address**.

d. Complete the Storefront Injection form.

Select where to display Apple Pay buttons on your site.

e. Complete the Payment Integration form.

- **Use Commerce Cloud Apple Pay Payment API?** Checked.
- **Payment Provider URL:**
 - Test: <https://apitest.visaacceptance.com/partner/demandware/payments/v1/authorizations>
 - Production: <https://api.visaacceptance.com/partner/demandware/payments/v1/authorizations>
- **Payment Provider Merchant ID:** Enter your Visa Acceptance Solutions merchant ID.
- **API Version:** v1.
- **Use Basic Authorization?** Unchecked.

- **Payment Provider User:** —
- **Payment Provider Password:** —
- **Use JWS?** Set to **Yes**.
- **JWS Private Key Alias:** Merchant.p12 Key Alias.

The private key alias is generated when you upload your .p12 key file (from Visa Acceptance Solutions self-serve) to Salesforce Business Manager under **Administration > Operations > Private Keys and Certificates**.

f. Click **Submit**.

2. Register your domain in Salesforce Business Manager.

a. Go to **Merchant Tools > Site Preferences > Apple Pay**.

b. Under the Domain Registration section, complete these fields:

- In the Apple Sandbox section, click **Register Apple Sandbox** to register Salesforce B2C to the Apple Sandbox account.
- In the Apple Production section, click **Register Apple Production** to register Salesforce B2C to the Apple Production account.

3. Set the transaction type.

Go to **Merchant Tools > Site Preferences > Custom Preferences > Apple Pay** and choose [Authorization](#) or [Sale](#).

Payer Authentication and 3-D Secure

Configure Payer Authentication and 3-D Secure for enhanced transaction security.



Note: *These settings apply to the Salesforce default card form (Direct API) only*

1. Go to **Merchant Tools > Site Preferences > Custom Preferences > Salesforce Default Acceptance Configuration**.
2. In **Payer Authentication** Mode, choose one of these options:
 - **Yes**: All transactions process with 3-D Secure.
 - **No**: No transactions process with 3-D Secure.
 - **Data Only + Yes**: Data Only is used for Visa and Mastercard/Maestro. All other card brands process with 3-D Secure.
 - **Data Only + No**: Data Only is used for Visa and Mastercard/Maestro. All other card brands process without 3-D Secure.
3. Enable the **Enable SCA** option to enforce Strong Customer Authentication.

Tokenization

Tokenization enables your customers to save their payment cards securely for future payments.

Follow these steps to enable tokenization:

1. Go to **Merchant Tools > Site Preferences > Custom Preferences > Tokenization Configuration**.
2. Enable the **Enable Tokenization Services** option to turn on Tokenization.
3. Enable the **Enable Limiting Saved Card** option to set the limits associated to saving cards.
4. In the **Saved Cards Allowed** option, enter the number of cards a customer can save in the defined time limit.
5. In the **Reset Interval** option, specify the number of hours before the saved card limit resets.
6. Enable the **Network Token Updates** option to prompt the cartridge to check for Token Life Cycle Updates prior to checkout. Your Visa Acceptance Solutions MID needs to be configured for Network Tokens.



Note: *The rate limiter settings for saving cards apply only to the Salesforce default card form (Direct API) and do not apply to Unified Checkout*

Fraud Screening

Enable Fraud Screening to alert the cartridge to look for fraud screening responses. Fraud Screening profiles must be set up in the Business Center.

Follow these steps to enable fraud screening:

1. Go to **Merchant Tools > Site Preferences > Custom Preferences > Salesforce Default Acceptance Configuration**.
2. Choose the **Enable Decision Manager Services** option. This setting applies for both Decision Manager and Fraud Management Essentials.

Updates to transactions marked as **REVIEW** are delivered via webhooks. The Update Job is available as a fallback option.

3. Set the **Conversion Detail Report Lookback Time** value.
If you are using REVIEW rules and configure the Decision Manager Update Job, set the number of hours to look back for updates to transactions in REVIEW status. The maximum value is 24 hours.
4. To enable the Decision Manager Update Job to poll for updates to the reviewed transactions, go to **Administration > Operations > Jobs** and select **Payment: Decision Manager Order Update**. Set these values:
 - **ID**: Enter a job ID.
 - **Description**: Enter the job description.
 - **ExecuteScriptModule.Module**: int_cybs_sfra_base/cartridge/scripts/jobs/DMOrderStatusUpdate.js
 - **ExecuteScriptModule.FunctionName**: orderStatusUpdate
 - **ExecuteScriptModule.Transactionnal**:
 - **True**: All changes occur as a single atomic operation. If any error occurs during the job, the system rolls back all changes to maintain data consistency.
 - **False**: No automatic rollback is applied. Merchants must handle transaction logic manually. This is often preferred for large batch jobs.
 - **ExecuteScriptModule.TimeoutInSeconds**: Set the function timeout value.

Device Fingerprint

Device fingerprinting collects information about the device used when paying for an order and can assist in fraud screening decisions.



Note: *The Device Fingerprint settings apply to the Salesforce default card form (Direct API) only.*

1. Go to **Merchant Tools > Site Preferences > Custom Preferences > Device Fingerprint Configuration**.
2. Enable the **Enable DeviceFingerprint Service** option.
3. In the **Organization ID** field, enter the Organization ID. Contact support if you do not know this value.
4. In the **ThreatMetrix URL** field, enter the URL that points to the JavaScript that generates and retrieves the fingerprint of the device.
5. In the **TTL (Time to Live)** field, enter the number of milliseconds to wait before generating a new fingerprint for any given customer session.

Delivery Address Verification

To verify the customer's shipping address during checkout, configure Delivery Address Verification services.

1. Go to **Merchant Tools > Site Preferences > Custom Preferences > Delivery Address Verification Configuration**.
2. Enable the **Delivery Address Verification Services** option.

Tax Calculation

To calculate local taxes once the customer enters their address at checkout, configure the Tax Calculation services.

1. Go to **Merchant Tools > Site Preferences > Custom Preferences > Tax Configuration** and set these options:
2. Enable the **Enable Tax Calculation** field.
3. Configure these tax settings:
 - **List of Nexus States:** List the states to calculate tax for.
 - **List of Nexus States to Exclude:** List the states to not calculate tax for.
 - **Merchants VAT Registration Number:** Enter your VAT registration number if you have one.
 - **Default Product Tax Code:** Enter the default tax code to use for products in the basket without a tax code.
 - **Purchase Order Acceptance City**

- **Purchase Order Acceptance State Code**
- **Purchase Order Acceptance Zip Code**
- **Purchase Order Acceptance Country Code**
- **Purchase Order Origin City**
- **Purchase Order Origin State Code**
- **Purchase Order Origin Zip Code**
- **Purchase Order Origin Country Code**
- **Ship From City**
- **Ship From State Code**
- **Ship From Zip Code**
- **Ship From Country Code**



Important: If you enable Tax Calculation and do not specify any states in the List of Nexus States or List of Nexus States to Exclude, Tax Calculation assumes every state or province is taxable. You can leave either the **List of Nexus States** or the **List of Nexus States to Exclude** as empty, but you cannot leave both empty.

Order Management

Salesforce B2C Commerce does not natively support order management functions. This cartridge has functions that can be used to process captures and authorization reversals.



Important: These functions must be customized before use in the Salesforce B2C Commerce user interface.



Note: *The ServiceFrameworkTest example controllers referenced below are for testing only. Access is granted only when **all** of the following are true:*

1. The Salesforce instance is **not** a production instance.
2. The **Enable Visa Acceptance Test Endpoints** preference (Custom Preferences > Visa Acceptance Cartridge configuration) is enabled. It is disabled by default.
3. The caller is an **authenticated, registered customer** who is a member of the `VisaAcceptanceTestAdmin` customer group

*The VisaAcceptanceTestAdmin customer group is not created by the metadata import — you must create it in Business Manager under **Merchant Tools > Customers > Customer Groups** and assign it only to authorized merchant staff. Only members of this group can access the test endpoints; any request from a non-member (or an unauthenticated visitor) is redirected to the home page.*

Capture

The capture function can be found in the script `scripts/http/capture.js`. A working example is available in the `ServiceFrameworkTest-TestCaptureService` controller.

Reference the `capture.js` object and make this request:

```
var captureObj = require("~/cartridge/scripts/http/capture.js");
var serviceResponse = captureObj.httpCapturePayment(requestID, merchantRefCode,
    paymentTotal, currency);
```

The resulting `serviceResponse` object contains the full response object generated by the request. The contents of this object determine your logic in handling errors and successes. These are the Capture request parameters:

Capture Request Parameters:

- `requestID`: The Visa Acceptance Solutions Request ID from the initial authorization.
- `merchantRefCode`: The Salesforce Order Number.
- `purchaseTotal`: The capture amount.
- `currency`: Currency Code.

Function Signature:

```
httpCapturePayment(requestID, merchantRefCode, purchaseTotal, currency)
```

Authorization Reversal

The authorization reversal function can be found in the script called *scripts/http/authReversal.js*. A working example is in the *ServiceFrameworkTest-TestAuthReversal* controller.

Reference the *AuthReversal.js* object and make this request:

```
var reversalObj = require("~/cartridge/scripts/http/authReversal.js");
var serviceResponse = reversalObj.httpAuthReversal(requestID, merchantRefCode,
paymentTotal, currency);
```

The resulting *serviceResponse* object contains the full response object generated by the request. The contents of this object determine your logic in handling errors and successes. These are the Authorization reversal request parameters:

Authorization Reversal Request Parameters:

- *requestID*: The Visa Acceptance Solutions Request ID from the initial authorization.
- *merchantRefCode*: The Salesforce Order Number.
- *purchaseTotal*: The reversal amount.
- *currency*: Currency Code.

Refund

The refund function can be found in the script called *scripts/http/refund.js*. A working example is in the *ServiceFrameworkTest-RefundService* controller.

Reference the *refund.js* object and make this request:

```
var refundObj = require("~/cartridge/scripts/http/refund.js");
var serviceResponse = refundObj.httpRefundPayment(transactionId, merchantRefCode,
paymentTotal, currency, refundEndpointType);
```

The resulting *serviceResponse* object contains the full response object generated by the request. The contents of this object determine your logic in handling errors and successes. These are the Authorization reversal request parameters:

Authorization Reversal Request Parameters:

- requestID: The Visa Acceptance Solutions Request ID from the capture or sale.
- merchantRefCode: The Salesforce Order Number.
- paymentTotal: The refund amount.
- currency: Currency Code.
- refundEndpointType: 'payments' for ACH/eCheck and 'captures' for all other payments

Function Signature:

```
httpRefundPayment(transactionId, referenceInformationCode, total, currency, refundEndpointType)
```



Note: Refunds are capped at the remaining refundable balance. A refund (whether full, a single partial, or the running total of multiple partials) that exceeds the captured amount is rejected before the gateway call.

Customization

The Visa Acceptance Solutions cartridge for Salesforce B2C Commerce has built-in custom hooks that can be used to customize the request data that is sent to each service.

These hooks can send additional custom data, such as Merchant Defined Data for authorization requests.

How Custom Hooks Work

After a request for a particular service is built, there is a check for any code registering to the hook `app.payment.modifyrequest`. If present, the hook is called for that specific request and the request object is passed into the hook. The return value of the hook is sent to Visa Acceptance Solutions as the final request object. Through this process, you can inject your own data into the request object from the custom code you write in a separate cartridge.

Implementation

To customize request objects, register the hook `app.payment.modifyrequest` in your cartridge's `hooks.json` file. An example would look like this, replacing the script path with your own script:

```
{
  "name": "app.payment.modifyrequest",
  "script": "./cartridge/scripts/hooks/modifyRequestExample"
}
```

You can copy the *scripts/hooks/modifyRequestExample* script from this cartridge into your own to use as a template for extending and modifying service request objects. Note that every hook must return a valid request object for the given service. Refer to the [Visa Acceptance Solutions REST API Field Reference](#) for information about any field you want to customize or add.

Support and Troubleshooting

Getting Support

If you require support with this extension, visit <https://support.visaacceptance.com> to raise a support case.

Required Information for Support Cases

Provide this information for your support case:

- Summary of the issue.
- Steps to reproduce the issue.
- Visa Acceptance Solutions B2C Commerce cartridge version.
- Visa Acceptance Solutions Merchant ID.
- Configuration screenshots: Provide screenshots of custom preference configurations.
- Log file and other relevant data: Download the logs from **Administration > Site Development > Development Setup > Log files**.

Upgrade

Upgrade the Visa Acceptance Solutions cartridge to a later version.

1. Download the code from the ISV Integration Toolkits section on [GitHub](#).
2. Zip the *payments_metadata* folder.
3. Go to **Administration > Site Development > Site Import & Export** and upload the *payments_metadata.zip* file.
4. Import the uploaded zip file.
5. Check release notes for any configuration parameters that might have changed.

The cartridge is successfully upgraded to the latest version.

Migrating to v2.0.0

Version 2.0.0 is a major release. When upgrading from an earlier release, review these changes:

1. **Add the new Business Manager cartridge.** Add `bm_cybs_sfra` to the Business Manager cartridge path (it provides the Webhook Manager page). See [Configuration \(on page 7\)](#).
2. Update the **service endpoints**. The endpoints have changed to `https://apitest.visaacceptance.com` (test) and `https://api.visaacceptance.com` (production).
3. Update the **Payment Credentials** service URL under **Administration > Operations > Services**.
4. **Re-check site preferences.** Preference groups were renamed to `VisaAcceptance_*` and regrouped. Payer Authentication, SCA, Decision Manager, and Transaction Type are now under **Salesforce Default Acceptance Configuration** and apply to the Salesforce default card form (Direct API) only. For Unified Checkout, configure these controls in the Visa Acceptance Solutions Business Center. Confirm your settings after importing metadata.
5. **Removed features:**
 - a. **Flex Microform** — card capture is now Unified Checkout or the Salesforce default payment form (Direct API).
 - b. **Standalone Google Pay** — Google Pay is now offered through Unified Checkout. Standalone Apple Pay is still supported.
 - c. **Network Token webhook** — token updates are now retrieved through the Token Management Service API.

6. **Set up webhooks.** Order updates are now driven by webhook notifications. Subscribe using the new Webhook Manager. The Salesforce jobs remain as a fallback. See [Webhooks \(on page 13\)](#).
7. **Authentication.** API authentication continues to use your REST Shared Secret and Key ID. Message-Level Encryption (MLE) uses a P12 certificate.
8. **Message Level Encryption.** The Message Level Encryption preference `VisaAcceptance_CertificateAlias` has been renamed `VisaAcceptance_RequestMLECertificateAlias`, and `VisaAcceptance_CertificateSerialNo` has been removed. New configuration steps are available for Request and Response MLE, including a master MLE switch (off by default).

Release Notes

Release history and updates for the Visa Acceptance Salesforce B2C Commerce cartridge.

Version 2.0.0 (September 2026)

Versioning changed from Calendar Versioning to Semantic Versioning

Enhancements:

- Renamed integration from Cybersource to Visa Acceptance Solutions
- Upgraded Unified Checkout to version 1.x, supporting multiple payment methods and services in a single integration
- Added Refund function
- Added webhook notifications for Fraud Screening and Unified Checkout events
- Added new Business Manager cartridge `bm_cybs_sfra`, including webhook manager page.
- Reorganized meta and site preference groupings
- Updated the authentication method to JSON web token
- Full support for Message Level Encryption
- Adjust Payer Authentication setup and Device Data Collection trigger (Direct API only)

Changes:

- Business Manager controls for Payer Authentication (including SCA), Decision Manager, and Transaction Type now only applies to the Salesforce default card form (direct API).
- Unified Checkout payment methods/services now configured in Business Center
- Google Pay now only supported through Unified Checkout
- Replaced Network Token lifecycle notifications with API based updates
- Added webhook notifications for Fraud Screening and Unified Checkout events
- Replaced Network Token lifecycle notifications with API based updates

Security:

- Addressed security issues

Removed:

- Microform

Bug Fix:

- Added Device Data Collection timeout handling

Version 26.2.0 (March 2026)

Enhancements:

- Updated Payer Authentication flow to align with SFRA best practices
- Added fallback device data capture for Payer Authentication
- Updated Mastercard 3-D Secure Data Only transactions
- Updated Cardinal Commerce URLs for Data Center Migration
- Modified subscription creation during the authorization call in checkout
- Decision Manager support for Apple Pay Transactions
- Apple Pay address handling improvements
 - Checkout page: collect address from checkout page
 - Cart/Mini Cart page: collect address from Apple Pay

Bug Fixes:

- Corrected page routing for failed Apple Pay scenarios
- Fetch the total amount from the server side instead of capturing it on the frontend to avoid issues for different currencies across different locales
- Correct handling of AUTHORIZED_RISK_DECLINED response for post authorization scenarios
- CheckoutServices Error Fix: Resolved TypeError occurring when the cartridge is disabled
- Fixed issue where the Unified Checkout capture context did not refresh when the Delivery Address Verification is enabled
- Corrected page redirection issue for Decision Manager reject scenarios
- Fixed bug causing "setAddress1" of null when performing follow-up transactions with a registered customer after updating shipping address.
- Fixed issue where the eCheck transient token exceeded the session.privacy 2000-character limit

Version 26.1.0 (January 2026)

New Feature:

- Added support for 3-D Secure Data Only transactions.

Version 25.4.0 (December 2025)

New Feature:

- Added support for Unified Checkout v0.32 (Card Payments, Apple Pay, Google Pay, Click to Pay, and eCheck).

Enhancement:

- End of support for Click to Pay legacy.

Version 25.3.0 (May 2025)

New Features:

- Added Payer Authentication support for Google Pay.
- Added multi-currency support for Google Pay.

Bug Fixes:

- Handled session variables in SCA flow.
- Removed encryption type from Microform v2 request.

Version 25.2.0 (March 2025)

New Feature:

- Message-Level Encryption (MLE).

Enhancement:

- Added support for Cartes Bancaires, Elo, China UnionPay, and JCB.

Version 25.1.0 (January 2025)

New Feature:

- Replaced Microform v0.11 with v2.

Bug Fixes:

- Added webhook subscription deletion if the subscription is deleted at Visa Acceptance Solutions or Salesforce custom object.
- Handled undefined exception scenario for 3-D Secure transactions.

Version 24.4.0 (September 2024)

New Feature:

- DMPA support.

Enhancement:

- Upgraded to jQuery v3.7.0.

Version 24.3.0 (August 2024)

Enhancements:

- Upgraded the cartridge to support SFRA v7.0.
- Added MOTO Commerce Indicator.

Version 24.2.1 (May 2024)

Bug Fixes:

- Checkmarx issues fixed.
- Device fingerprint bug fixed.

Version 24.2.0 (April 2024)

New Features:

- Network token support

Enhancements:

- Implemented Direct API integration for Payer Authentication, adding Payer Authentication Setup and Device Data Collection.
- Enhanced Strong Consumer Authentication (SCA).

Version 24.1.0 (February 2024)

New Features:

- Added Strong Customer Authentication retries for card payments.

Enhancements:

- SFRA v6.3 support.
- Salesforce B2C Commerce Release 22.7 support.
- Renamed Visa SRC to Click to Pay.
- Implemented Sale functionality for Credit Card, Google Pay, Click to Pay and Apple Pay.
- Updated flex script referring from v0.11.0 to v0.11.
- Updated API header in Http Signature Authentication.

Version 21.1.0 (June 2021)

Enhancements:

- Improved Payer Authentication screen (modal).

Bug Fixes:

- Added descriptive error messages on certain fail cases and invalid inputs.
- Reloading on the final confirmation page does not result in a failed authorization.

Version 20.2.0 (February 2021)**New Features:**

- Google Pay
- Visa Secure Remote Commerce payment method
- Improved the security on the My Account page by adding Microform to tokenize payment cards.

Bug Fixes:

- Improved the security of keys by changing data type of password fields from `String` to `password`.
- Added more security to the exposed parameters of device fingerprint.

Version 20.1.1 (November 2020)**Bug Fixes:**

- Improved the security on accessing and modifying sensitive fulfillment-related actions on an order (for example, order acceptance, canceling etc.).

Version 20.1.0 (August 2020)**Initial release supporting:**

- Credit/debit cards
- Apple Pay
- Payer Authentication/3-D Secure
- Delivery Address Verification service
- Tax Calculation service
- Authorization, Capture, Authorization Reversal

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